

How Will I Explain OOP?

The examples in the first few chapters use text-based Python; these sample programs get input from the user and output information to the user purely in the form of text. I'll introduce OOP by showing you how to develop text-based simulations of physical objects in code. We'll start by creating representations of light switches, dimmer switches, and TV remote controls as objects. I'll then show you how we can use OOP to simulate bank accounts and a bank that controls many accounts.

Once we've covered the basics of OOP, I'll introduce the *pygame* module, which allows programmers to write games and applications that use a *graphical user interface (GUI)*. With GUI-based programs, the user intuitively interacts with buttons, checkboxes, text input and output fields, and other user-friendly widgets.

I chose to use pygame with Python because this combination allows me to demonstrate OOP concepts in a highly visual way using elements on the screen. Pygame is extremely portable and runs on nearly every platform and operating system. All the sample programs that use the pygame package have been tested with the recently released pygame version 2.0.

I've created a package called `pygwidgets` that works with pygame and implements a number of basic widgets, all of which are built using an OOP approach. I'll introduce this package later in the book, providing sample code you can run and experiment with. This approach will allow you to see real, practical examples of key object-oriented concepts, while incorporating these techniques to produce fun, playable games. I'll also introduce my `pyghelpers` package, which provides code to help write more complicated games and applications.

All the example code shown in the book is available as a single download from the No Starch website: <https://www.nostarch.com/object-oriented-python/>.

The code is also available on a chapter-by-chapter basis from my GitHub repository: <https://github.com/IrvKalb/Object-Oriented-Python-Code/>.